6.1 Properties of Exponents

Name _____ Period ____ Date _____

Algebra I 6.1 · Student sheet

I can…

- use zero and negative exponents.
- use the properties of exponents to simplify expressions.
- use the properties of exponents with scientific notation.

Vocabulary · fill in the missing word

Zero exponent — a^0 = _____

Negative exponent — a^{-n} = _____

Product of powers — $a^m \cdot a^n$ = _____

Quotient of powers — a^m/a^n = _____

Power of a power — $(a^m)^n$ = _____

Power of a product / quotient — $(ab)^m$ = _____ / $\left(\frac{a}{b}\right)^m = a^m/b^m$

Part 1 · Pattern down

Going down the table, divide by 2 each time. Keep the pattern going.

Power	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}
Value	16	8	4	2				

So a^0 = _____ and a^{-n} = _____

Write each as a product of x's, then count: $x^2 \cdot x^3$ = _____ = $x^{_}$ x^5/x^2 = _____ = $x^{_}$

Part 2 · Notes: fill in the blanks

a^0 = _____ a^{-n} = _____ $(a \neq 0)$

$a^m \cdot a^n$ = _____ a^m/a^n = _____ $(a^m)^n$ = _____

$(ab)^m$ = _____ $(a/b)^m$ = _____

An expression is simplified when it has no _____ exponents and each base appears _____.

Part 3 · Examples (work them with the class)

Example 1

Evaluate each expression.

a. 6.7^0 b. $(-3)^{-4}$ c. $\frac{5^0}{2^{-2}}$

Example 2

Simplify. Write your answer using only positive exponents.

a. $4x^{-3}$ b. $\frac{1}{2b^{-5}}$

Example 3

Simplify. Write your answer using only positive exponents.

a. $x^3 \cdot x^{-5}$ b. $\frac{w^{10}}{w^4}$ c. $(z^4)^{-3}$

Example 4

Simplify.

a. $(-1.5y)^2$ b. $\left(\frac{a}{-10}\right)^3$ c. $\left(\frac{3d}{2}\right)^4$

Example 5

Evaluate. Write your answer in scientific notation and in standard form.

a. $(4.1 \times 10^{-3})(2 \times 10^7)$ b. $\frac{8.4 \cdot 10^6}{2.1 \cdot 10^2}$

Part 4 · Partner practice

Partner 1

Simplify. Write your answer using only positive exponents.

$(2x^3y^{-2})^{-3}$

Partner 2

Evaluate. Write your answer in scientific notation and in standard form.

$(6 \times 10^{5})(3 \times 10^{-2})$